1. FAMILY TREE

AIM

To write a Prolog program to implement a Family Tree using facts and rules to determine relationships such as father, mother, grandparent, and sibling.

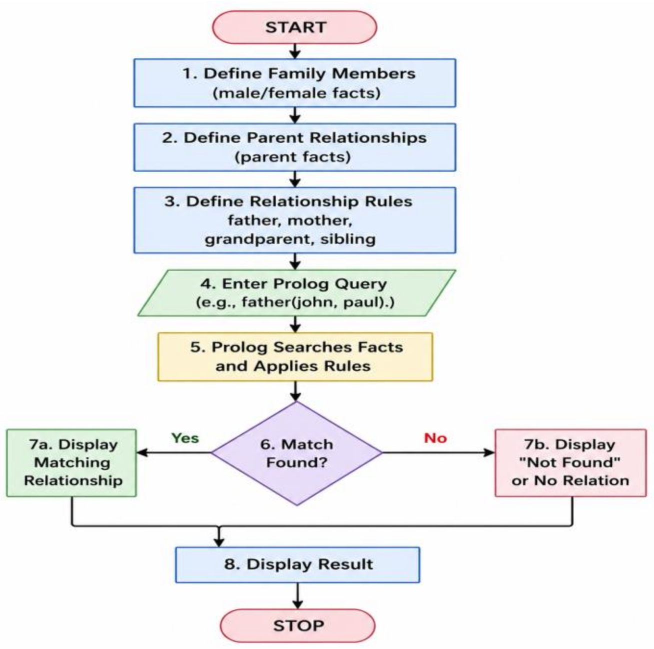
OBJECTIVE

- To represent family relationships using Prolog facts.
- To define rules for different family relationships.
- To query the database and find relationships between family members.
- To understand the use of logical inference in Prolog.

ALGORITHM

1. Start.
2. Define male and female family members.
3. Define parent relationships using facts.
4. Define rules for father, mother, grandparent, and sibling.
5. Enter a query for the required relationship.
6. Prolog searches the facts and applies the rules.
7. Display the matching family relationship.
8. Stop.

FLOWCHART



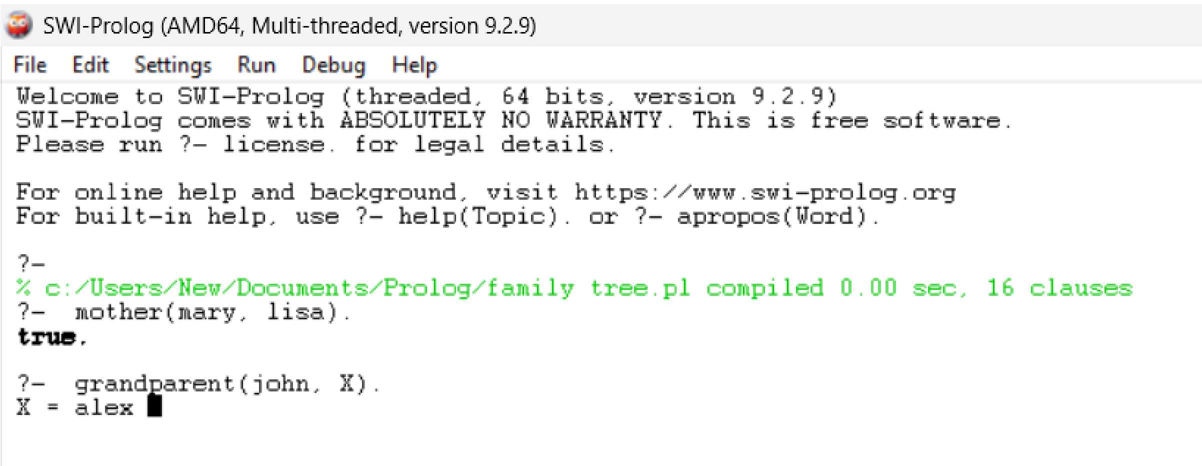
CODE:

```
parent(john, mary).

parent(john, david).

parent(mary, alex).

parent(david, lisa).

mother(mary, lisa).

grandparent(X, Y) :-
    parent(X, Z),
    parent(Z, Y).
```

QUERY

?- mother(mary, lisa).

OUTPUT



```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File  Edit  Settings  Run  Debug  Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/New/Documents/Prolog/family tree.pl compiled 0.00 sec, 16 clauses
?-  mother(mary, lisa).
true.

?-  grandparent(john, X).
X = alex
```

RESULT

Thus, the Family Tree was successfully implemented in Prolog, and family relationships such as father, mother, grandparent, and sibling were successfully determined using facts, rules, and queries.